

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

20 minutes · 20 points



Reference

Bayes' rule	$P(A \mid B) = P(B \mid A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B \mid A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathcal{H}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Name: Mateo SilvaStudent ID: 002243864**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

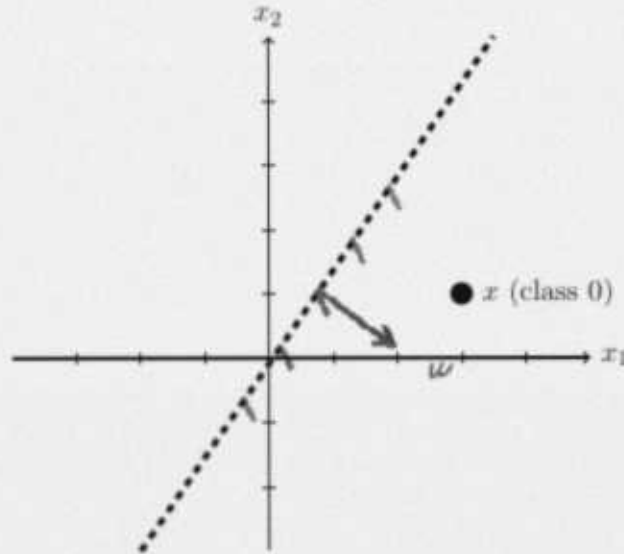
$$\begin{aligned}P(L|K=2) &= P(K=2|L) P(L) / P(K=2) \\P(K=2|L) &= .384 \quad P(K=2|F) = .375 \\P(K=2) &= P(K=2|L)P(L) + P(K=2|F)P(F) \\&= .384(.4) + .375(.6) \\&= .1536 + .225 \\&= .3786 \\P(L|K=2) &= .1536 / .3786 \\&\sim 0.406\end{aligned}$$

hmm

units?

Name: Matteo SilvaStudent ID: 002243864**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\begin{aligned}
 \text{slope} &= (5 - (-3)) / (3.5 - (-2)) \\
 &= 8 / 5.5 \sim 1.45 \\
 x_2 + 3 &= 1.45 (x_1 + 2) \\
 x_2 &= 1.45 x_1 - 0.09 \\
 w &= [1.45, -1], b = -0.09 \\
 x = [3, 1]: & 4.36 - 1 - .09 > 0 \\
 (b) \text{ no, any } c > 0 & \text{ scales } (w, b) \\
 w &= [-1.45, 1], b = 0.09
 \end{aligned}$$

Name: Mateo SilvaStudent ID: 002243854**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$\begin{aligned} z &= 0.8(3) - 2 = 0.4 \\ &= 1 / 1.670 \\ P(\text{pass} | x=3) &\sim 0.60 \end{aligned}$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$\begin{aligned} \sigma(0.4) &= 0.4 \\ p &= .231, .401, .769, .881 \\ &= .263 + .513 + .263 + .127 \\ L &\sim 1.17 \end{aligned}$$

Name: Mateo Silva

Student ID: 002245864

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

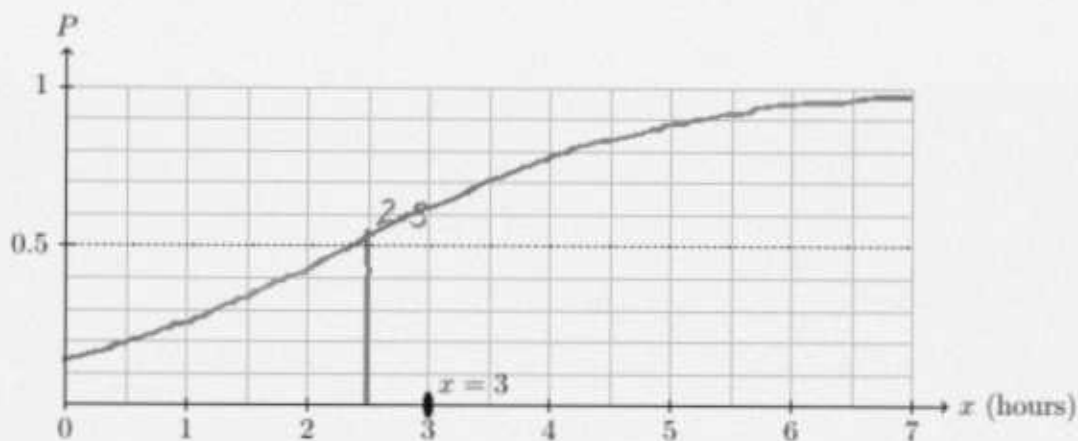
$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5) \\ = .231 + .802 - .924 - .595 = -.486$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

$$w = .8 - .1(-.486) = .849$$

$$b = -2 - .1(.282) = -2.03$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$
 $x = 2 / 0.8 = 2.5 \text{ hours}$

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Reference

Bayes' rule	$P(A B) = P(B A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Q3 cont.

$L \sim 2.00$

$$z = 0.5(4) - 1.5 = 0.5$$

$$p = .269, .378, .5, .731, .878$$

$$3.5/9 = .389$$

check

~~$w^\top x + b < 0?$~~

$$e^{-0.4} = .670$$

$$.6 \times .375 = .225$$

$$.1029 / .2904$$

$$8/5.5 = 1.4545$$

Name: Ben CarterStudent ID: 0057 01307**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

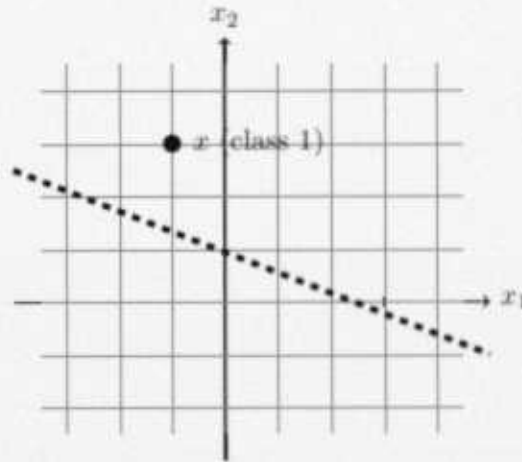
k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$\begin{aligned}
 P(H|K=3) &= P(K=3|H) P(H) / P(K=3) \\
 P(K=3) &= P(K=3|H)P(H) + P(K=3|G)P(G) \\
 &= .4116(.25) + .25(.75) \\
 &= .1029 + .1875 \\
 &= .2904 \\
 P(H|K=3) &= .1029 / .2904 \\
 &\sim 0.354 \\
 .6 \times .375 &= .225 \\
 .1029 / .2904
 \end{aligned}$$

Name: Ben CarterStudent ID: 005701301**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$= -3.5 / 9 \sim -0.39$$

$$x_2 = -0.39 x_1 + 0.94$$

$$x = [-1, 3]^T: -0.39 + 3 = 2.61$$

(b) not unique: $c w, c b, c > 0$

$$w = [0.39, 1], b = -0.94$$

$$8/5.5 = 1.4545$$

$$2 \times 0.378 = 0.756$$

$$1/2.49 = 0.401$$

Name: Ben CarterStudent ID: 005701307**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$\sigma(0.5) = 1 / (1 + e^{-0.5})$$

$$= 1 / 1.607$$

$$P(\text{pass} | x=4) = 0.62$$

$$3.5/9 = .389$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

$$p = .269, .378, .5, .731, .818$$

$$\sigma(0.4) = 0.4$$

$$L = -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)]$$

$$L = 2.00$$

$$3.5/9 = .389$$

see scratch →

hmm

Name: Ben CarterStudent ID: 005701301

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$dL/dw = -269 + .378(2) - .5(3) - .269(5) - .182(6)$$

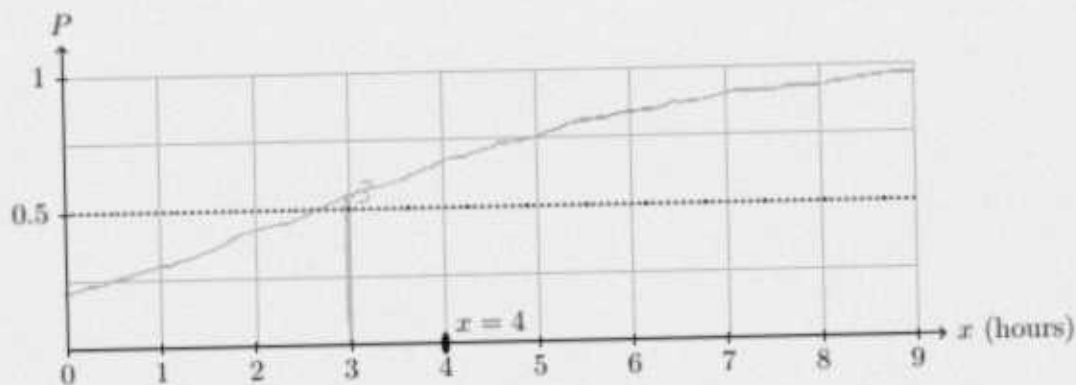
$$= -269 + .756 - 1.5 - 1.345 - 1.092 = -2.91$$

$$dL/db = -269 + .378 - .5 - .269 - .182 = -3.04$$

$$w = .5 - .05(-2.91) = .646$$

$$b = 1.5 - .05(-3.04) = 1.48$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$P = .5 \text{ where } z = 0$$

$$x = 1.5 / .5 = 3 \text{ hours}$$

Reference

Bayes' rule	$P(A B) = P(B A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathcal{H}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1/(1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Q3 cont.

$$.384 \times .4 = .1536$$

$$1/2.49 = .401$$

$$\begin{aligned} d\mathcal{L}/dw &= .231(1) + .401(2) - .231(4) - .119(5) \\ &= .231 + .802 - .924 - .595 = -.486 \end{aligned}$$

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Name: Maya TorresStudent ID: 004363484**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L)P(L) / P(K=2)$$

$$P(K=2|L) = .384 \quad P(K=2|F) = .375$$

$$P(K=2) = P(K=2|L)P(L) + P(K=2|F)P(F)$$

$$= .1536 + .225$$

$$= .3786$$

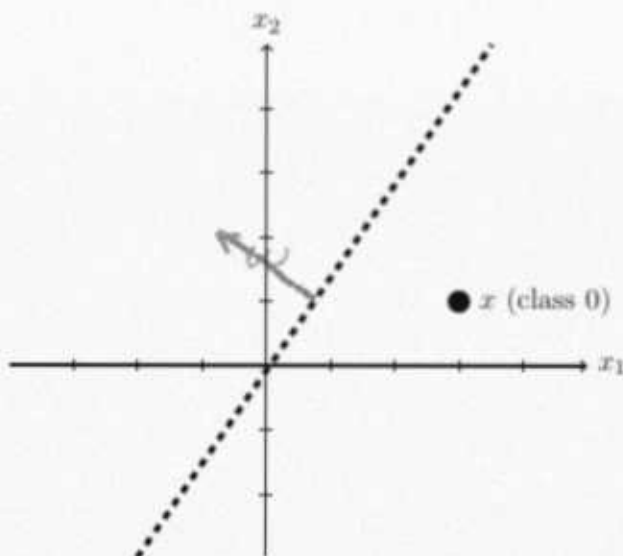
$$= .406$$

$$.384 \times .4 = .1536$$

check

Name: Maya TorresStudent ID: 004563484**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\begin{aligned} \text{slope} &= (5 - (-3)) / (3.5 - (-2)) \\ &= 8 / 5.5 \approx 1.45 \\ 1.2 + 3 &= 1.45 (4 + 2) \\ 4.2 &= 1.45 (4) - 0.09 \end{aligned}$$

and

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

→ class 1, wrong side! flip

(b) no, any $c > 0$ scales (w, b)

$$w = [-1.45, 1], b = 0.09$$

$$2 \times .378 = .756$$

Name: Maya TorresStudent ID: 004563484**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$\bar{z} = 0.8(3) - 2 = 0.4$$

$$\text{sigma}(0.4) = 1 / (1 + e^{-0.4})$$

$$\text{sigma}(0.4) = 0.4$$

$$= 1 / 1.670$$

$$P(\text{pass} | x=3) = 0.60$$

$$3.5/9 = .389$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$\bar{z} = -1.2, -0.4, 1.2, 2$$

$$p = .23, .40, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L = 1.17$$

Name: Maya TorresStudent ID: 004563484

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .23(1) + .40(2) - .23(4) - .119(5)$$

$$= .231 + .802 - .924 - .595 = -.486$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

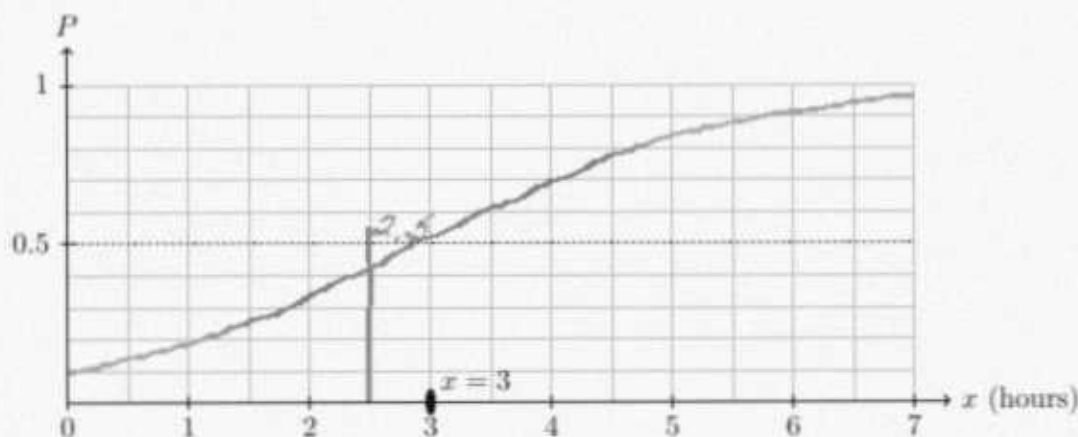
$$w = .8 - .1(-.486) = .849$$

$$b = 2 - .1(.282) = 1.9718$$

$$\ln .5 = -.693$$

$$1 + 1.49 = 2.49$$

- (d) Sketch $P(\text{pass} | x)$ under the original (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

$$.384 \times .4 = .1536$$

see scratch →

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Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

$$3.5/9 = .389$$

Name: Ethan ParkStudent ID: 00442297**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

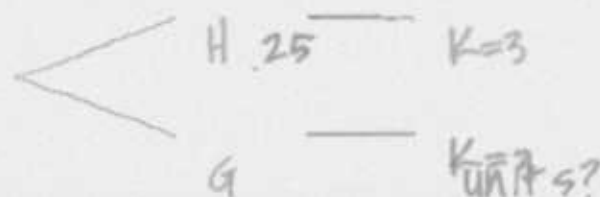
$$= .4116(.25) + .25(.75)$$

$$= .1029 + .1875$$

$$= .2904$$

$$P(H|K=3) = .1029 / .2904$$

$$0.354$$



Name: _____

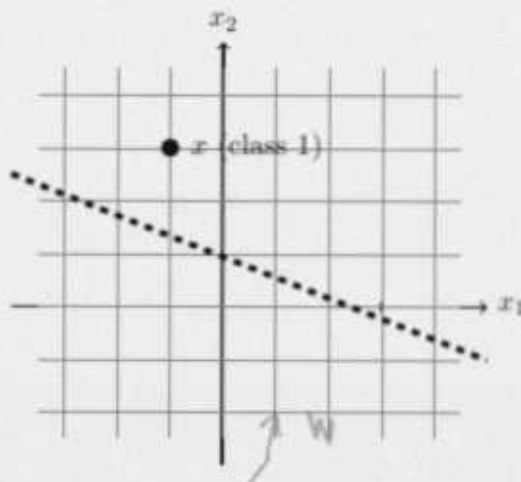
Student ID: _____

Ethan Park

004422197

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}\{w^T x + b > 0\}$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$w = [1, 1.45]$$

$$= -3.5 / 9 \sim -0.39$$

$$x_2 = -0.39 x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 = 2.61$$

$$= 1.67 > 0 \rightarrow \text{class 1 ok}$$

(b) not unique: $c w, c b, c > 0$

$$w = [0.39, 1], b = -0.94$$

$$4 \times 375 = 225$$

$$8/5.5 = 1.4545$$

$$35/9 = 3.89$$

check



Name: Ethan PARKStudent ID: 004422197**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$\begin{aligned}
 z &= 0.5(4) - 1.5 = 0.5 \\
 &= 1 / 1.607 \\
 P(\text{pass} | x=4) &\approx 0.62 \\
 6 \times .375 &= .225 \\
 85.5 &= 14545
 \end{aligned}$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$\begin{aligned}
 z &= -1, -5, 0, 1, 1.5 \\
 p &= .269, .378, .5, .731, .818 \\
 L &= -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)] \\
 &= .313 + .475 + .693 + .313 + .201 \\
 L &\approx 2.00
 \end{aligned}$$

redo

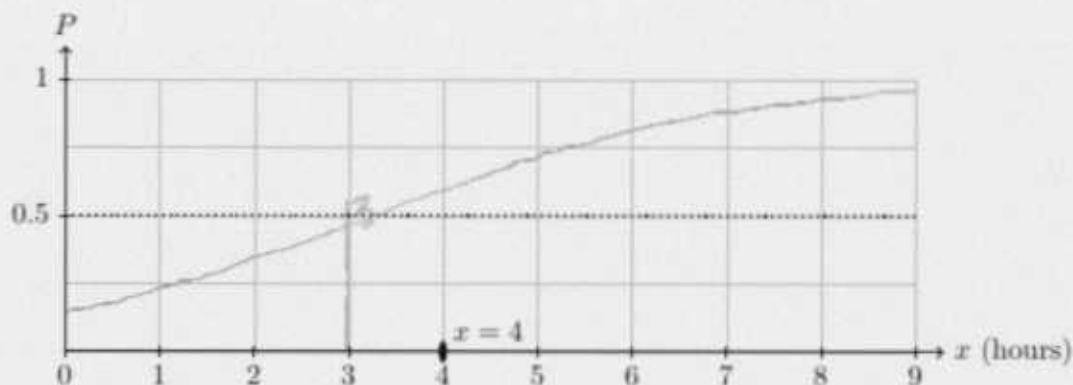
Name: Ethan ParkStudent ID: 004422197

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$\begin{aligned}
 &= .269 + .756 - 1.5 - 1.345 = -1.092 \approx -2.91 \\
 dL/db &= .269 + .378 - .5 - .269 - .182 = -.304 \\
 w &= .5 - .05(-2.91) = .646 \\
 b &= -1.5 - .05(-.304) = -1.48
 \end{aligned}$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$x = 1.5 / .5 = 3 \text{ hours}$$

$$1029 / 2904$$

$$\ln .5 = -.693$$

$$= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91$$

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Scratch work (not graded)

$$8/5.5 = 1.4545$$

$$2 \times .378 = .756$$

Name: Froya LarsenStudent ID: 001218018**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

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You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .4116(.25) + .25(.75)$$

$$= .1029 + .1875$$

$$= .2904$$

$$P(H|K=3) = .4116 / .2904$$

$$P(H|K=3) = .1029 / .2904$$

$$\sim 0.354$$

$$6 \times .375 = 2.25$$

$$1 + 1.49 = 2.49$$

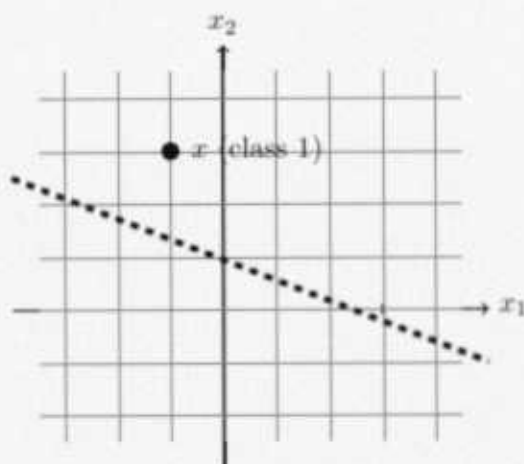
$$\sqrt{2.49} = .401$$

Name: Freya Larsen

Student ID: 00218018

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$= -3.5 / 9 \sim -0.39$$

$$x_2 = -0.39 x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 = 2.61$$

$$= 1.67 > 0 \rightarrow \text{class 1 OK}$$

(b) not unique: $c \cdot w, c \cdot b, c > 0$

$$w = [0.39, 1] \quad b = -0.94$$

$$384 \times 4 = 1536$$

x2?

Name: Froya LarsenStudent ID: 00278078**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$h(0) ?$
 $z = 0.5(4) - 1.5 = 0.5$
 $\text{sigma}(0.5) = 1 / (1 + e^{-0.5})$
 $= 1 / 1.607$
 $P(\text{pass} | x=4) \sim 0.62$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$z = -1.5, 0, 1, 1.5$
 $p = .269, .378, .5, .731, .818$
 $p = e^z$
 $L = -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)]$
 $= 3.13 + 4.75 + .693 + 3.13 + 2.01$
 $L \sim 13.72$
 $b \times 3.75 = 1.875$
 $\ln .5 = -.693$

see below

Name: Freyd LarsenStudent ID: 00178078

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$dL/dw = .269 + .378(2) - .5(3) - .269(5) - .182(6)$$

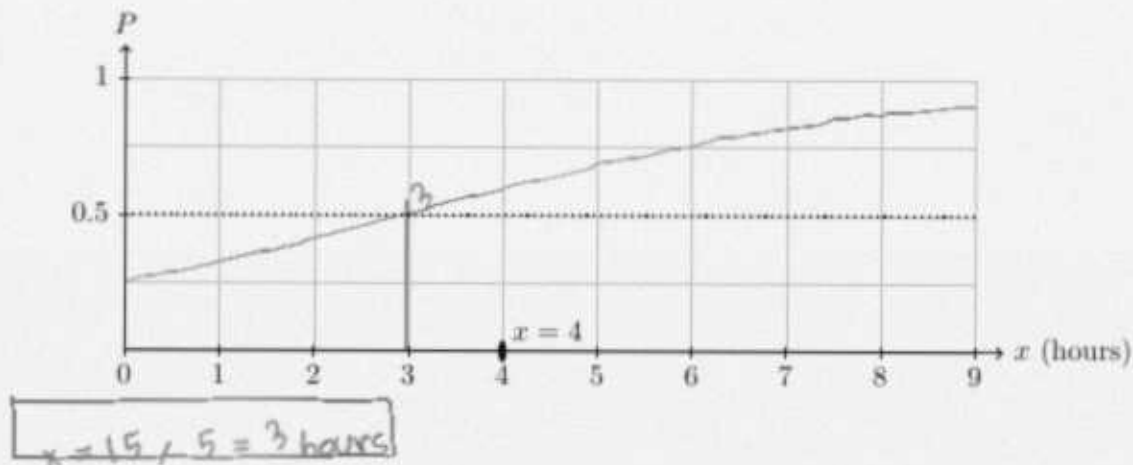
$$= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.9$$

$$dL/db = .269 + .378 - .5 - .269 - .182 = -.304$$

$$w = .5 - .05(-2.9) = .646$$

$$b = -.5 - .05(-.304) = -.48$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$1 + 1.49 = 2.49$$

$$e^{-0.4} = .670$$

$$2 \times .378 = .756$$

redo

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

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- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

20 minutes · 20 points

Reference

Bayes' rule	$P(A \mid B) = P(B \mid A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B \mid A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Name: Amara OseiStudent ID: 002160217**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\sigma(0.4) = 1 / (1 + e^{-0.4})$$

$$= 1 / 1.670$$

$$P(\text{pass} | x=3) \sim 0.60$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

Name: Amaria Osei

Student ID: 002160217

Problem 3 (continued)

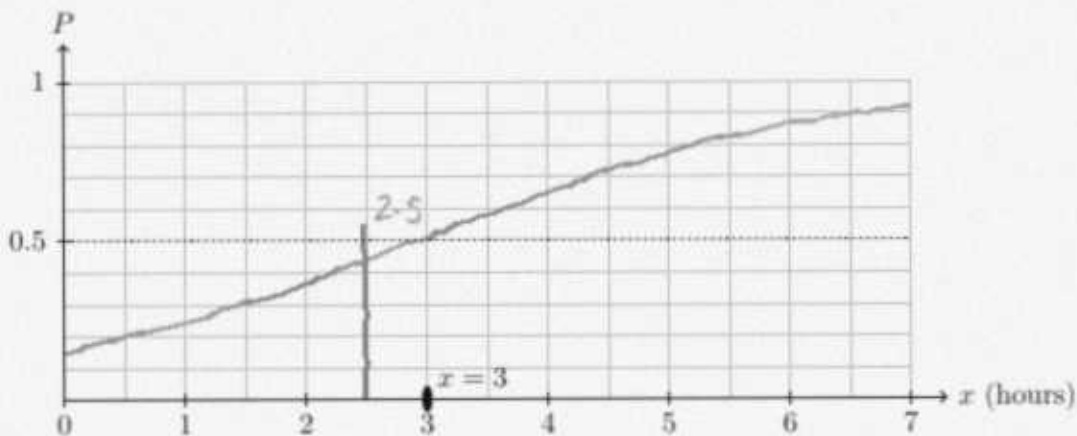
- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$\begin{aligned} dL/dw &= .231(1) + .401(2) - .231(4) - .119(5) \\ &= .231 + .802 - .924 - .595 = -.486 \end{aligned}$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

$$\begin{aligned} w &= .8 - .1(-.486) = .849 \\ b &= -2 - .1(.282) = -2.03 \end{aligned}$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



Switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

$$3.5 / 9 = .389$$

$$1 / 2.49 = .401$$



Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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Reference

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Total probability	$P(B) = \sum_i P(B \mid A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Name: Yuki TanakaStudent ID: 005516324**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

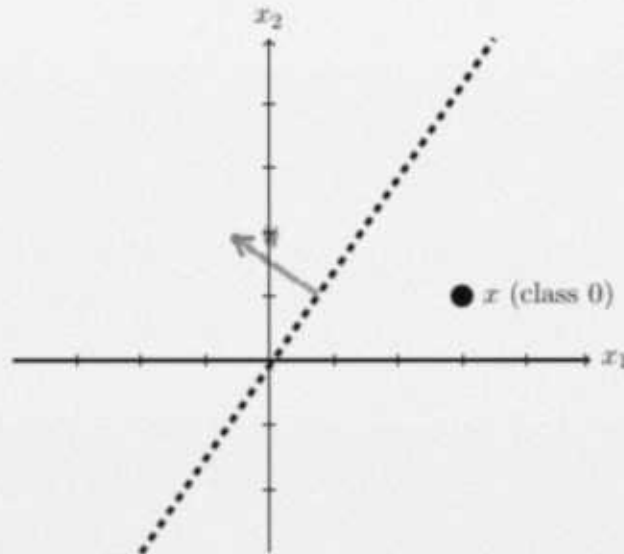
$$\begin{aligned}
 P(L|K=2) &= P(K=2|L)P(L) / P(K=2) \\
 P(K=2|L) &= 0.384 \quad P(K=2|F) = 0.375 \\
 P(K=2) &= P(K=2|L)P(L) + P(K=2|F)P(F) \\
 &= 0.384(0.4) + 0.375(0.6) \\
 &= 0.1536 + 0.225 \\
 &= 0.3786 \\
 P(L|K=2) &= 0.1536 / 0.3786 \\
 &= 0.406
 \end{aligned}$$

$$\begin{aligned}
 3.59 &= 389 \\
 1 + 1.49 &= 2.49 \\
 1/2.49 &= .401
 \end{aligned}$$

?

Name: Yuki TanakaStudent ID: 005516324**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2)) \\ = 8 / 5.5 = 1.45$$

$$x_2 + 3 = 1.45(x_1 + 2)$$

$$x_2 = 1.45x_1 - 0.09$$

$$w = [1.45, -1] \quad b = -0.09$$

→ class 1, wrong side! Flip
 (b) No any $c > 0$ scales (1, -1)
 $w = [-1.45, 1] \quad b = 0.09$

q?

Name: Yufei TanakaStudent ID: 005516324**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\sigma(0.4) = 1 / (1 + e^{-0.4})$$

$$= 1 / 1.670$$

$$P(\text{pass} | x=3) = 0.60$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = 1.11$$

Name: Yuki TanakaStudent ID: 005516324**Problem 3 (continued)**

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$\begin{aligned} dL/dw &= 23(1) + 40(2) - 23(4) - 119(5) \\ &= 231 + 802 - 924 - 595 = -486 \end{aligned}$$

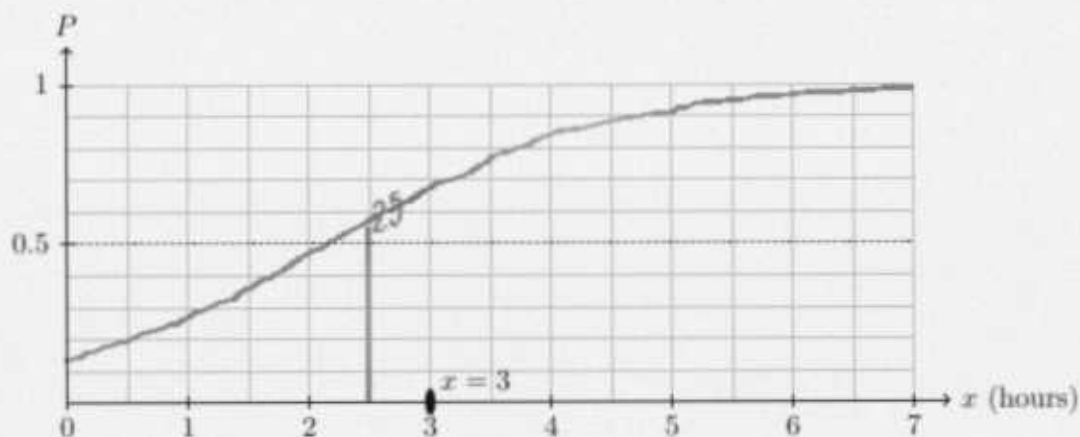
$$w = 8 - 1(-486) = 849$$

$$b = -2 - 1(282) = -284$$

$$384 \times 4 = 1536$$

$$e^{-0.4} = 0.670$$

- (d) Sketch $P(\text{pass} \mid x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



Switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

units?

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

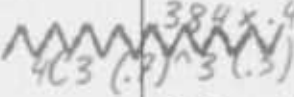
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- Calculators are allowed. No notes, no phones.

Reference

Bayes' rule	$P(A B) = P(B A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Q1 cont.


$$384 \times .4 = .1536$$

$$4C3 (.7)^3 (.3)$$

$$P(K=2/L) = .384 \quad P(K=2/F) = .375$$

$$1/2.49 = .401$$

Name: Zara AliStudent ID: 007902592**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$\begin{aligned}
 P(L|K=2) &= P(K=2|L) P(L) / P(K=2) \\
 P(K=2) &= P(K=2|L)P(L) + P(K=2|F)P(F) \\
 &= .384(.4) + .375(.6) \\
 &= .1536 + .225 \\
 &= .3786
 \end{aligned}$$

$$\begin{aligned}
 P(L|K=2) &= .1536 / .3786 \\
 &= 0.406
 \end{aligned}$$

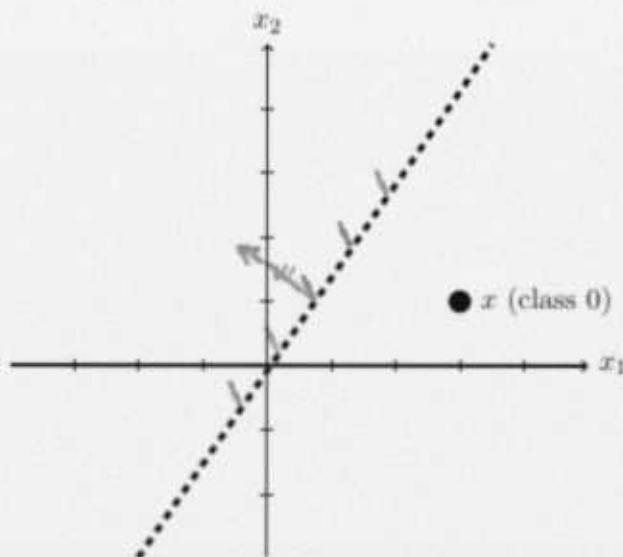
$$1 + 1.49 = 2.49$$

$$.1029 / .2904$$

see scratch $\rightarrow$

Name: Zara AliStudent ID: 007902592**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.



$$\text{slope} = (5 - (-3)) / (3.5 - (-2)) = 8 / 5.5 \sim 1.45$$

hmm

$$x_2 + 3 = 1.45 (x_1 - 2)$$

$$x_2 = 1.45 x_1 - 0.09$$

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - 0.09 > 0$$

$\rightarrow$ class 1, wrong side! flip

(b) no, any $c > 0$ scales (w, b)

$$w = [-1.45, 1], b = 0.09$$

Name: Zara AliStudent ID: 007902592**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?ln(0)??

$$z = 0.8(3) - 2 = 0.4$$

$$\sigma(0.4) = 1 / (1 + e^{-0.4})$$

$$= 1 / 1.670$$

$$P(\text{pass} | x=3) \sim 0.60$$

$$1 + 1.49 = 2.49$$

see below

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

!

Name: Zara AliStudent ID: 007902592**Problem 3 (continued)**

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5) \quad \text{units!}$$

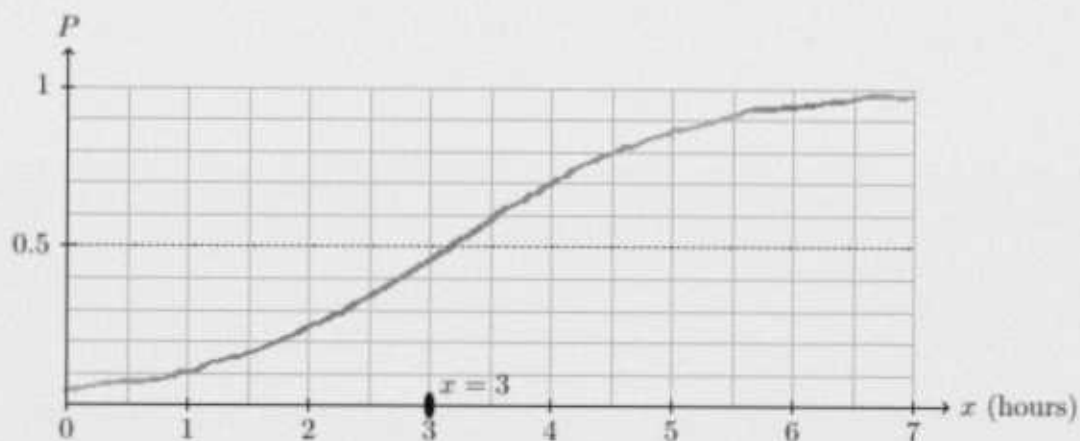
$$= .231 + .802 - .924 - .595 = -.486$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

$$w = .8 - .1(-.486) = .849$$

$$b = -2 - .1(.282) = -2.03$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



Switch where $wx + b = 0$

$$x = -b/w = 2/0.8 = 2.5 \text{ hours}$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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20 minutes · 20 points

Reference

Bayes' rule	$P(A B) = P(B A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Name: Lucas MeyerStudent ID: 008437697**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

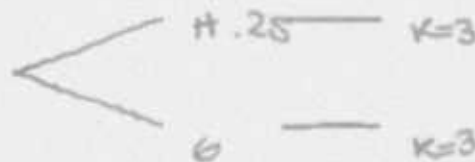
$$\begin{aligned}
 P(K=3|H) &= .4116 & P(K=3|G) &= .25 \\
 &= .4116(.25) + .25(.75) \\
 &= .1029 + .1875 \\
 &= .2904
 \end{aligned}$$

$$P(H|K=3) = .1029 / .2904$$

$$= 0.354$$

$$1/2.49 = .401$$

$$e^{-0.4} = .670$$

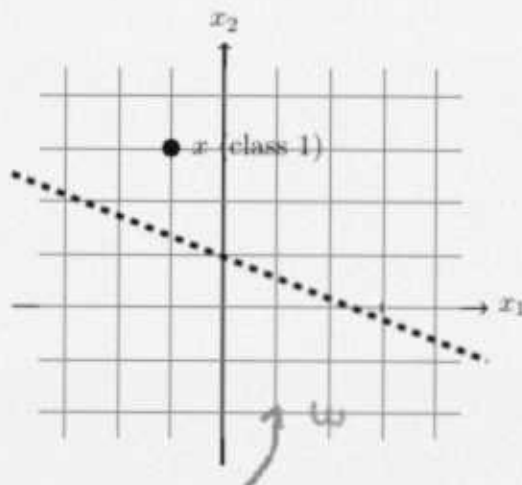


Name: Lucas Meyer

Student ID: 008477697

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

~~3~~ 3

$$\text{slope} = (-1 - 2.5) / (5 - (-4)) \\ = -3.5 / 9 \approx -0.39$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x_2 = -0.39x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 - .94$$

$$= 1.67 > 0 \rightarrow \text{class 1 ok}$$

(b) not unique: $cw, cb, c > 0$

$$w = [0.39, 1], b = -0.94$$

hmm
hmm

Name: Lucas MeyerStudent ID: 008437697**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$\sigma(0.5) = 1 / (1 + e^{-0.5})$$

$$= 1 / 1.607$$

$$P(\text{pass} | x=4) \sim 0.62$$

$$1 + 1.49 = 2.49$$

$$\ln .5 = -.693$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -.5, 0, 1, 1.5$$

$$p = .269, .378, .5, .731, .818$$

$$L = -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)]$$

$$L \sim 2.00$$

$$2 \times .378 = .756$$

$$\ln .5 = -.693$$

Name: Lucas MeyerStudent ID: 008477677

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$= .269 + .756 - 1.5 - 1.345 \quad \text{~~2.91~~$$

$$dL/db = .269 + .378 - .5 - .269 - .182 = -.304$$

$$w = .5 - .05(-2.91) = .646$$

$$b = 1.5 - .05(-.304) = 1.48$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$p = .5 \text{ where } z = 0$$

$$x = 1.5 / .5 = 3 \text{ hours}$$

$$3.5/9 = .389$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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Total probability	$P(B) = \sum_i P(B A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Name: Tessa QuinnStudent ID: 00517868**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3|H) = .416 \quad P(K=3|G) = .25$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .416(.25) + .25(.75)$$

$$= .1029 + .1875$$

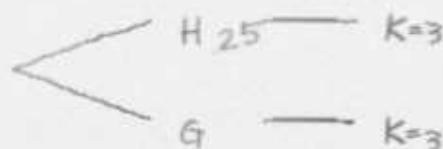
$$= .2904$$

$$P(H|K=3) = .1029 / .2904$$

$$\sim 0.354$$

$$.35 / .9 = .389$$

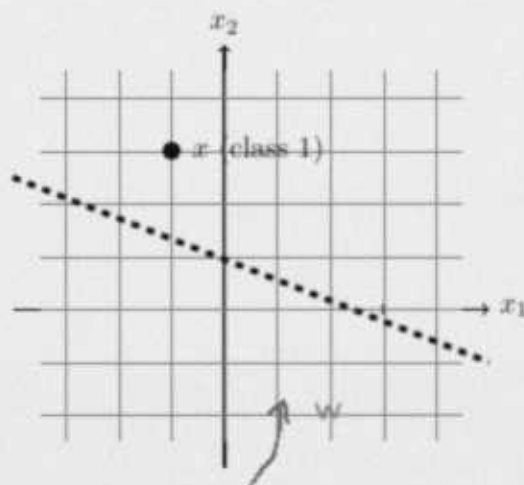
$$.1029 / .2904$$



check

Name: Tessa QuinnStudent ID: 00517860**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{1}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (-1 - 2.5) / (5 - (-4))$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x_2 = -0.39x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 = .94$$

$$= 1.67 > 0 \rightarrow \text{class 1 ok}$$

(b) not unique: $c w, c b, c > 0$

$$w = [0.39, 1], b = -0.94$$

$$2 \times .378 = .756$$

$x_2?$

Name: Tessa Quinn

Student ID: 005178601

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$\frac{dL}{dw} = 269 + 378(2) - 5(3) - 269(5) - 182(6)$$

$$x = 0.5$$

$$= 269 + 756 - 15 - 1345 - 1092$$

$$dL/db = 269 + 378 - 5 - 269 - 182 = -304$$

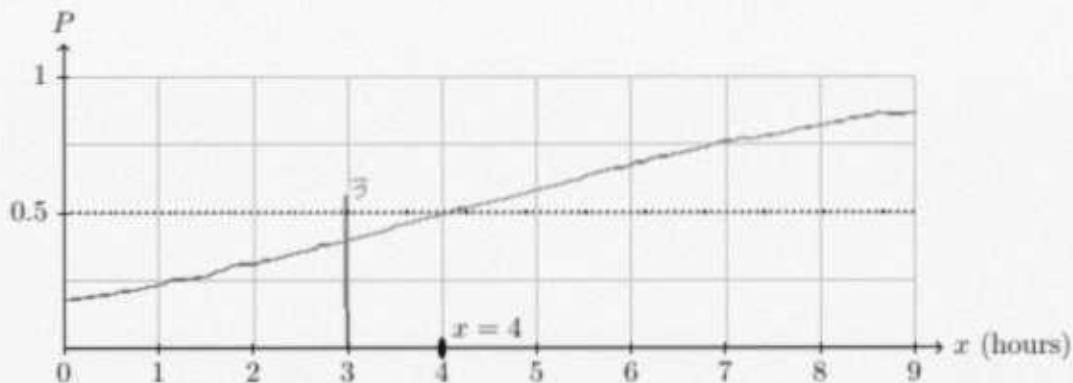
$$w = 5 - 0.05(-2.7) = 6.46$$

$$b = -15 - 0.05(-304) = -14.8$$

see below

units?

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$P = 0.5 \text{ where } z = 0$$

$$x = 15 / 5 = 3 \text{ hours}$$

$$\ln 5 = -0.693$$

$$35/9 = 3.89$$

Name: Tessa QuinnStudent ID: 0057869**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$= 1 / 1.607$$

$$P(\text{pass} | x=4) \sim 0.62$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.5, 0, 1.5$$

$$L = -[\ln(.73) + \ln(.62) + \ln(.5) + \ln(.73) + \ln(.818)]$$

$$= .33 + .475 + .693 + .33 + .20$$

$$L \sim 2.00$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

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- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
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Reference

Bayes' rule	$P(A \mid B) = P(B \mid A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B \mid A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1/(1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Name: guth AbabeStudent ID: 002014276**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$= .1029 + .1875$$

$$P(H|K=3) = .1029 / .2904$$

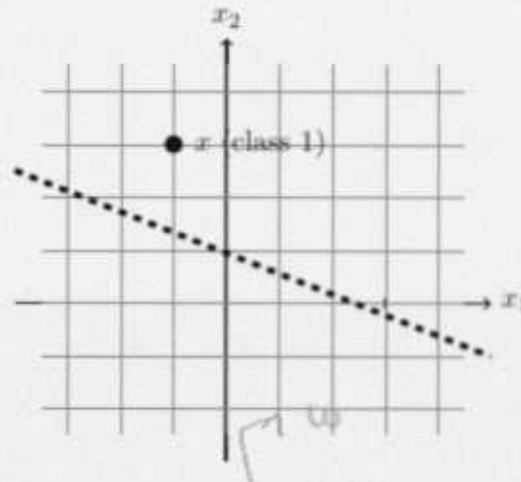
$$\sim 0.354$$

$$1 + 1.49 = 2.49$$



Name: auth abebeStudent ID: 002014246**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathcal{H}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$= -3.5 / 9 = -0.39$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x_2 = -0.39x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 = 2.61$$

$$= 2.61 > 0 \rightarrow \text{class 1 OK}$$

(b) not unique, $cw, cb, c > 0$

$$w = [0.39, 1], b = -0.94$$

$$0/5.5 = 1.4545$$

check

Name: Ruth AbebeStudent ID: 002014276**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$\sigma(0.5) = 1 / (1 + e^{-0.5})$$

$$= 1 / 1.607$$

$$P(\text{pass} | x=4) = 0.62$$



ans

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

$$p = .269, .378, .5, .731, .818$$

$$L = -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)]$$

$$= .313 + .475 + .693 + .313 + .201$$

$$L = 2.00$$

←

Name: Ruth AbebeStudent ID: 002014276**Problem 3 (continued)**

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$\frac{dL}{dw} = .269 + .378(2) - .5(3) - .269(5) - .182(6)$$

$$\frac{dL}{db} = .269 + .378 - .5 - .269 - .182 = -.304$$

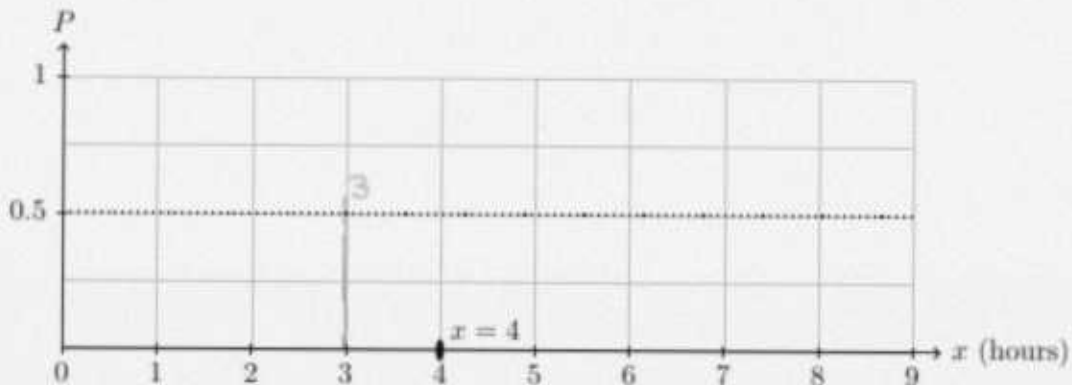
$$w = .5 - .05(-2.91) = .646$$

$$b = -1.5 - .05(-.304) = -1.48$$

$$3.59 = .389$$

$$1/2.49 = .401$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$P = .5 \text{ where } z = 0$$

$$x = 1.5 / .5 = 3 \text{ hours}$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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Total probability	$P(B) = \sum_i P(B A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Q1 cont.

$$P(H|K=3) = .1029 / .2904$$

$$= .4116(.25) + .25(.75)$$

~ 0.354

~~$$P(L|K=2) = .384$$~~

$$= .1029 + .1875$$

units?

$$1/2 \cdot .49 = .401$$

$$.384 \times .4 = .1536$$

$$\ln .5 = -.693$$

hmm

Name: _____

Student ID: _____

Clara Rossi

001781787

Problem 1 [10 pts] Which coin?

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .4116(.25) + .25(.75)$$

$$= .1029 + .1875$$

$$P(H|K=3) = .1029 / .2904$$

$$\sim 0.354$$

$$\frac{.1029}{.2904} = .354$$

see scratch 

redo

ans

Name: _____

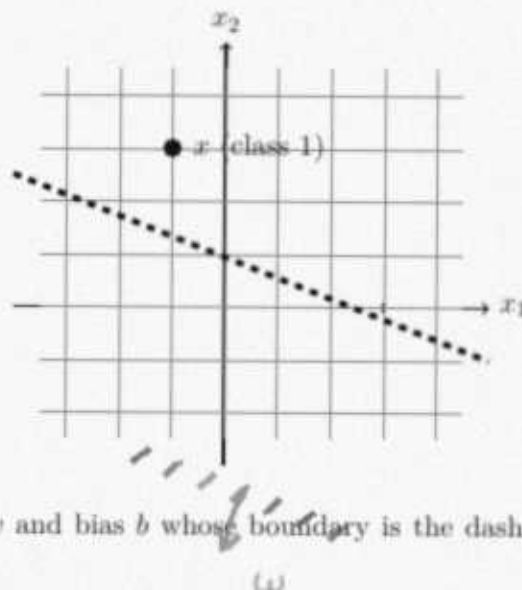
Clara Rossi

Student ID: _____

001781787

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^\top x + b > 0]$. The plotted point $x = [-1, 3]^\top$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.

2. Is your (w, b) the only one that works? Explain.

$$w = [9, 3.5]$$

$$\text{slope} = (-1 - 2.5) / (5 - (-4)) \\ = -3.5 / 9 \sim -0.39$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x_2 = -0.39x_1 + 0.94$$

$$= 1.67 > 0 \rightarrow \text{class 1 ok}$$

(b) not unique $cw, cb, c > 0$

$$w = [0.39, 1], b = -0.94$$

$$8/5.5 = 1.4545$$

$$1/2.49 = .401$$

units?

Name: Clara RossiStudent ID: 001781787**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$\sigma(0.5) = 1 / (1 + e^{-0.5})$$

$$= 1 / 1.607$$

$$p = e^z$$

$$P(\text{pass} | x=4) \sim 0.67$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

$$p = .269, .378, .5, .731, .818$$

$$L = -[\ln(.731) + \ln(.622) + \ln(.5) + \ln(.731) + \ln(.818)]$$

$$= .313 + .475 + .693 + .313 + .201$$

$$L \sim 2.00$$

$$3.5/9 = .389$$

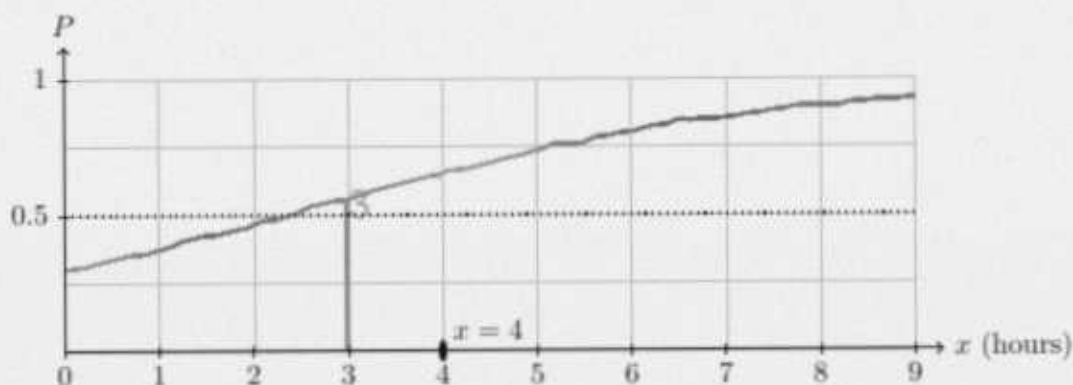
Name: Clara RossiStudent ID: 001781787**Problem 3 (continued)**

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$\begin{aligned} &= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91 \\ dL/db &= .269 + .378 - .5 - .269 - .182 = -.304 \end{aligned}$$

$$\begin{aligned} w &= .5 - .05(-2.91) = .646 \\ b &= -1.5 - .05(-.304) = -1.48 \end{aligned}$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$$\begin{aligned} P &= .5 \text{ where } z = 0 \\ x &= 1.5 / .5 = 3 \text{ hours} \\ 3.5/9 &= .389 \end{aligned}$$

x2?

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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Total probability $P(B) = \sum_i P(B | A_i) P(A_i)$

Binomial $P(K = k) = \binom{n}{k} p^k (1-p)^{n-k}$

Linear classifier $\hat{y} = \mathbb{1}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$

Sigmoid $\sigma(z) = 1/(1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Q1 cont.

$$2 \times .378 = .756$$

$$.384 \times .4 = .1536$$

$$3.5/9 = .389$$

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$1 + 1.49 = 2.49$$

$$8/5.5 = 1.4545$$

$$e^{-0.4} = .670$$

$$3.5/9 = .389$$

$$.384 \times .4 = .1536$$

!!

redo

Name: Owen GallagherStudent ID: 003957047**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$P(K=2|L) = .384 \quad P(K=2|F) = .375$$

$$P(K=2) = P(K=2|L)P(L) + P(K=2|F)P(F)$$

$$= .1536 + .225$$

$$= .3786$$

$$= .384(.6) + .375(.4)$$

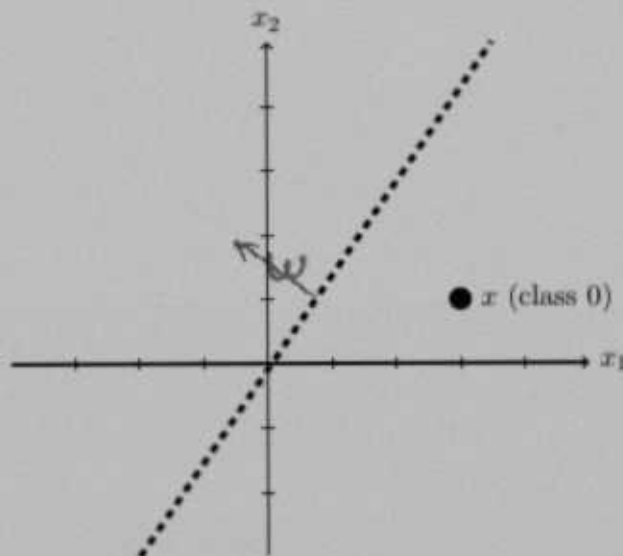
$$P(L|K=2) = .1536 / .3786$$

$$\sim 0.406$$

$$1/2.49 = .401$$

Name: Owen GallagherStudent ID: 003957047**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$= 8 / 5.5 \sim 1.45$$

$$x_2 + 3 = 1.45(x_1 + 2)$$

$$x_2 = 1.45x_1 - 0.09$$

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

→ class 1, wrong side! flip

(b) no, any $c > 0$ scales (w, b)

$$w = [-1.45, 1], b = 0.09$$

$$.6 \times .375 = .225$$

$$1 + 1.49 = 2.49$$



!!

!!

Name: Owen GallagherStudent ID: 003957047**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\sigma(0.4) = 1 / (1 + e^{-0.4})$$

$$= 1 / 1.670$$

$$P(\text{pass} | x=3) = 0.60$$

$$\ln .5 = -.693$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1.2, -0.4, 1.2, 2$$

$$p = .231, .401, .769, .881$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

hmm

x2?

Name: Owen GallagherStudent ID: 003957047

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$\begin{aligned} dL/dw &= .231(1) + .401(2) - .231(4) - .119(5) \\ &= .231 + .802 - .924 - .595 = -.486 \end{aligned}$$

$$dL/db = .231 + .401 - .231 - .119 = .282$$

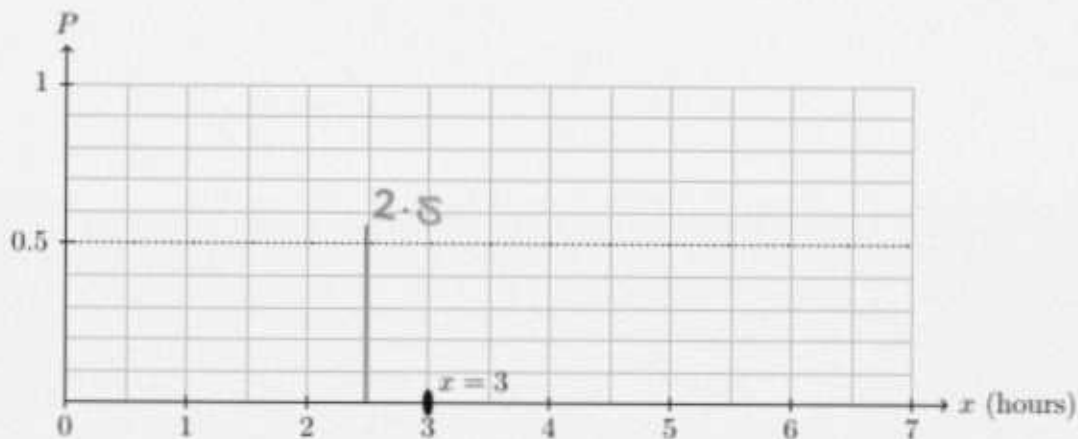
$$w = 8 - .1(-.486) = 8.49$$

$$b = -2 - .1(.282) = -2.03$$

$$\ln .5 = -.693$$

$$1/2.49 = .401$$

- (d) Sketch $P(\text{pass} \mid x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$

$$2 \times .378 = .756$$

$$.384 \times .4 = .1536$$

$$e^{-0.4} = .670$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

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Reference

Bayes' rule	$P(A B) = P(B A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

1029/2904

Name: Nikhil RaoStudent ID: 005730535**Problem 1 [10 pts] Which coin?**

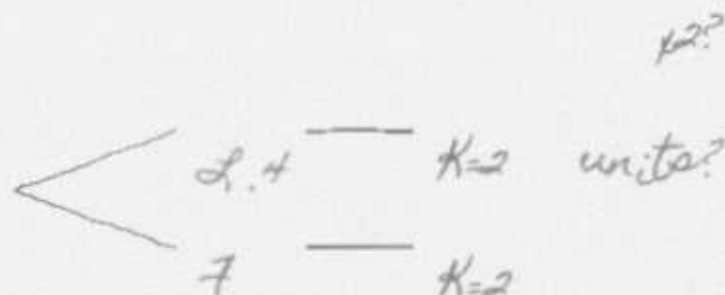
A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

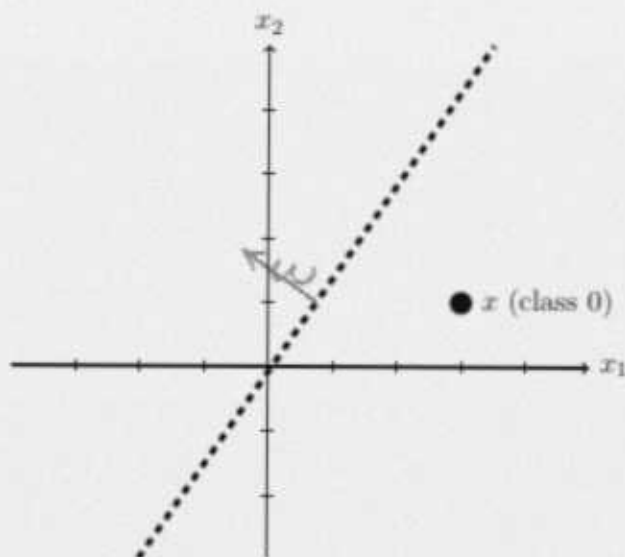
$$\begin{aligned}
 P(L|K=2) &= P(K=2|L) P(L) / P(K=2) \\
 P(K=2|L) &= .384 \quad P(K=2|F) = .375 \\
 P(K=2) &= P(K=2|L)P(L) + P(K=2|F)P(F) \\
 &= .384(.4) + .375(.6) \\
 &= .1536 + .225 \\
 &= .3786
 \end{aligned}$$

$$\boxed{P(L|K=2)} = .1536 / .3786$$



Name: Nikhil RaoStudent ID: 005730535**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}\{w^\top x + b > 0\}$. The plotted point $x = [3, 1]^\top$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$= 8/5.5 \sim 1.45$$

$$x^2 + 3 = 1.45(x + 2)$$

$$x^2 = 1.45x - 0.09$$

$$x = [3, 1]: 4.36 - 1 - .09 > 0$$

$\rightarrow$ class 1, wrong side! flip

$$w = [-1.45, 1], b = 0.09$$

$$3.5/9 = .389$$

check

Name: Nikhil RaoStudent ID: 005730535**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?~~ln(0.4)?~~

$$z = 0.8(3) - 2 = 0.4$$

$$\sigma(0.4) = 1 / (1 + e^{-0.4})$$

$$P(\text{pass} | x=3) = 0.60$$

$$2 \times .378 = .756$$

$$1 + 1.49 = 2.49$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$\vec{z} = -1.2, -0.4, 1.2, 2$$

$$L = -[\ln(.769) + \ln(.599) + \ln(.769) + \ln(.881)]$$

$$= .263 + .513 + .263 + .127$$

$$L \sim 1.17$$

$$8/5.5 = 1.4545$$

Name: Nikhil RaoStudent ID: 005730535

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

$$dL/dw = .231(1) + .401(2) - .231(4) - .119(5) \\ = .231 + .802 - .924 - .595 = -.486$$

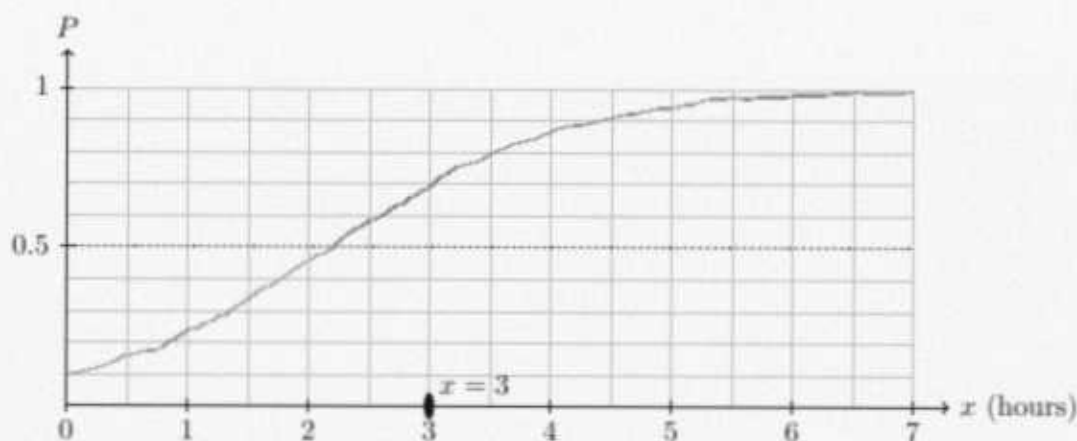
$$dL/db = .231 + .401 - .231 - .119 = .282$$

$$w = .8 - .1(-.486) = .849$$

$$b = -2 - .1(.282) = -2.03$$

$$.1029/.2904$$

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $wx + b = 0$

$$x = -2 / 0.8 = -2.5 \text{ hours}$$

$$1/2.49 = .401$$

$$8/5.5 = 1.4545$$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

- Write your name and student ID at the top of every question page.
- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Reference

Bayes' rule	$P(A \mid B) = P(B \mid A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B \mid A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Name: Leila FarahStudent ID: 007187191**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin G is fair; coin H is loaded and lands heads with probability 0.7. You pick coin H with probability 0.25, flip it four times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.0625	0.0081
1	0.2500	0.0756
2	0.3750	0.2646
3	0.2500	0.4116
4	0.0625	0.2401

You see $K = 3$ heads. What is the probability that you picked the loaded coin?

$$P(H|K=3) = P(K=3|H) P(H) / P(K=3)$$

$$P(K=3|H) = .4116 \quad P(K=3|G) = .25$$

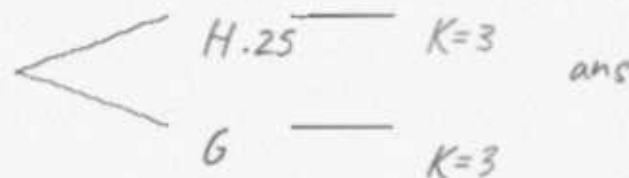
$$P(K=3) = P(K=3|H)P(H) + P(K=3|G)P(G)$$

$$= .4116(.25) + .25(.75)$$

$$= .2904$$

$$\sim 0.354$$

$$.6 \times .375 = .225$$

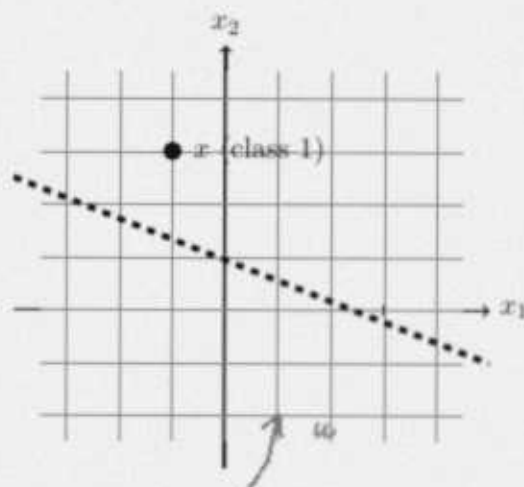


Name: Leila Farah

Student ID: 007187191

Problem 2 [10 pts] Draw the boundary

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}[w^T x + b > 0]$. The plotted point $x = [-1, 3]^T$ belongs to class 1.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$w \cdot x + b < 0?$$

$$= -3.5 / 9 \sim -0.39$$

$$x_2 - 2.5 = -0.39(x_1 + 4)$$

$$x_2 = -0.39x_1 + 0.94$$

$$x = [-1, 3]: -0.39 + 3 = .94$$

$$= 1.67 > 0 \rightarrow \text{class 1 ok}$$

(b) not unique: a w, c b, c > 0

$$w = [0.39, 1], b = -0.94$$

$$1/2 \cdot 49 = .401$$

$$2 \times .378 = .756$$

$$e^{-0.4} = .670$$

Name: Leila FarahStudent ID: 007187791**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.5$ and $b = -1.5$. The training set is

student	hours x_i	passed y_i
1	1	0
2	2	0
3	3	1
4	5	1
5	6	1

(a) What does the model give for $P(\text{pass} | x = 4)$?

$$z = 0.5(4) - 1.5 = 0.5$$

$$\sigma(0.5) = 1 / (1 + e^{-0.5})$$

$$P(\text{pass} | x=4) = 0.62$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$z = -1, -0.5, 0, 1, 1.5$$

$$p = .269, .378, .5, .731, .818$$

$$L = .313 + .475 + .693 + .313 + .207$$

$$L = 2.00$$

Name: Leila Farah

Student ID: 007187191

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.05$: give the new w and b .

$$dL/dw = .269 + .378(2) - .5(3) - .269(5) - .182(6)$$

$$= .269 + .756 - 1.5 - 1.345 - 1.092 = -2.91$$

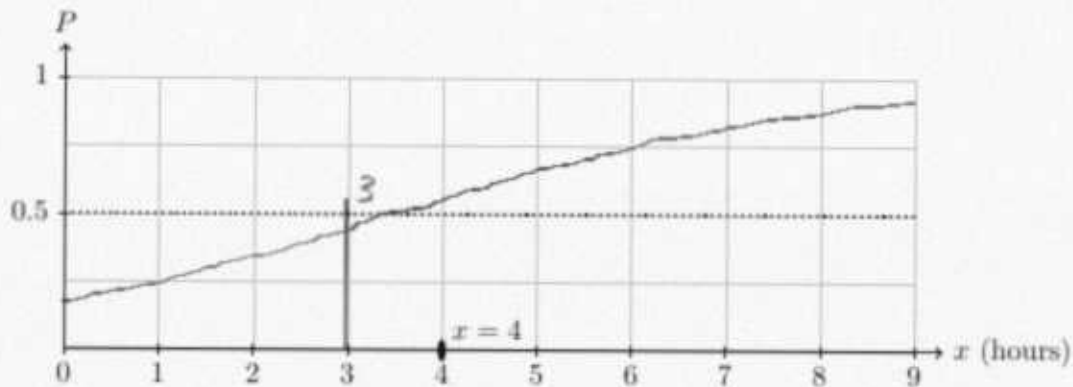
$$w = .5 - .05(-2.91) = .646$$

$$b = -1.5 - .05(-.304) = -1.48$$

$$e^{-0.4} = .670$$

units?

- (d) Sketch $P(\text{pass} | x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



$P = .5 \text{ where } z = 0$
 $x = 4.5 / .5 = 3 \text{ hours}$

Quiz 1

Probability and linear classifiers
(fictional course, for the scan_split example)

Instructions

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- Keep all work for a question on that question's page; use its back if you need more room.
- Show your work; answers without work earn partial credit only.
- Calculators are allowed. No notes, no phones.

Reference

Bayes' rule	$P(A B) = P(B A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathcal{H}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1/(1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)

Q1 cont.

$$= .1536 + .225$$

$$.384 \times .4 = .1536$$

$$.6 \times .375 = .225$$

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$1 + 1.49 = 2.49$$

$$8/5.5 = 1.4545$$

$$= .3786$$

Name: Innas BergStudent ID: 009908762**Problem 1 [10 pts] Which coin?**

A drawer holds two coins. Coin F is fair; coin L is loaded and lands heads with probability 0.8. You pick coin L with probability 0.4, flip it three times, and record the number of heads K .

k	$P(K = k \mid \text{fair})$	$P(K = k \mid \text{loaded})$
0	0.125	0.008
1	0.375	0.096
2	0.375	0.384
3	0.125	0.512

You see $K = 2$ heads. What is the probability that you picked the loaded coin?

$$P(L|K=2) = P(K=2|L) P(L) / P(K=2)$$

$$P(K=2|L) = .384 \quad P(K=2|F) = .375$$

$$P(K=2) = P(K=2|L)P(L) + P(K=2|F)P(F)$$

$$= .384(.4) + .375(.6)$$

$$= .1536 + .225$$

$$= .3786$$

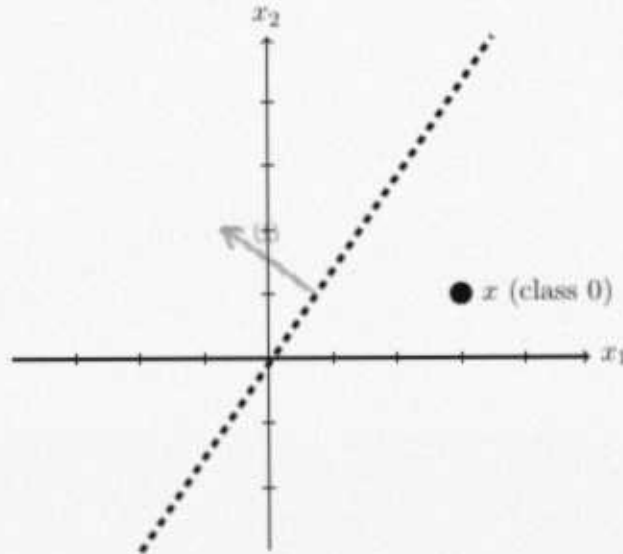
$$P(L|K=2) = .1536 / .3786$$

$$\sim 0.406$$

see scratch $\rightarrow$

Name: Jonas BergStudent ID: 009908762**Problem 2 [10 pts] Draw the boundary**

The dashed line is the decision boundary we want from a linear classifier $\hat{y} = \mathbb{I}\{w^T x + b > 0\}$. The plotted point $x = [3, 1]^T$ belongs to class 0.



1. Give a weight vector w and bias b whose boundary is the dashed line and which put x in its class.
2. Is your (w, b) the only one that works? Explain.

$$\text{slope} = (5 - (-3)) / (3.5 - (-2))$$

$$w = [9, 3.5]$$

$$x_2 + 3 = 1.45(x_1 + 2)$$

$$x_2 = 1.45x_1 - 0.09$$

$$w = [1.45, -1], b = -0.09$$

$$x = [3, 1]: 4.36 - 1 = 3.36 > 0$$

→ class 1, wrong side! flip

(b) no, any $c > 0$ scales (w, b)

$$w = [-1.45, 1], b = 0.09$$

Name: Jonas BergStudent ID: 009908762**Problem 3 [10 pts] Pass or fail**

(two pages)

A logistic model predicts whether a student passes from the hours x they studied:

$$P(\text{pass} | x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

with current parameters $w = 0.8$ and $b = -2$. The training set is

student	1	2	3	4
hours x_i	1	2	4	5
passed y_i	0	0	1	1

(a) What does the model give for $P(\text{pass} | x = 3)$?

$$z = 0.8(3) - 2 = 0.4$$

$$\text{sigma}(0.4) = 1 / (1 + e^{-0.4})$$

$$= 1 / 1.670$$

~~m(0)??~~

$$P(\text{pass} | x=3) \sim 0.60$$

$$3.5/9 = .389$$

$$1079/2904$$

(b) Compute the log-loss $L = -\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, where $p_i = \sigma(wx_i + b)$.

$$p = .231, .401, .769, .881$$

$$= .263 + .513 + .263 + .127$$

x2?

$$L = .231 + .401 + .769$$

$$L \sim 1.17$$

Name: Jonas Berg

Student ID: 009908762

Problem 3 (continued)

- (c) The gradient is $\partial L / \partial w = \sum_i (p_i - y_i) x_i$ and $\partial L / \partial b = \sum_i (p_i - y_i)$. Take one gradient step with learning rate $\eta = 0.1$: give the new w and b .

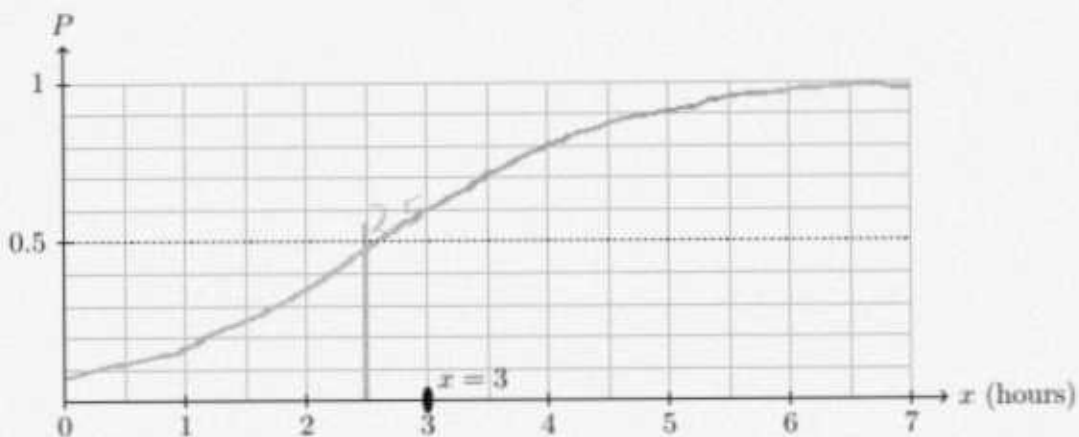
$$= .231 + .802 - .924 - .595 = -.486$$

~~$$b = -2 + .03$$~~

$$w = .8 - .1(-.486) = .849$$

$$b = -2 - .1(2.82) = -2.03$$

- (d) Sketch $P(\text{pass} \mid x)$ under the *original* (w, b) on the axes below, and mark the hours at which the model's prediction switches to pass.



switch where $w x + b = 0$

$$x = 2 / 0.8 = 2.5 \text{ hours}$$